

Rajveer Sodhi

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Education

Brown University, Sc.M Computer Science (GPA: 4.00)

Sep 2025 - May 2027

University of British Columbia Okanagan, B.Sc Computer Science Honors (GPA: 3.82)

Sep 2021 - May 2025

Awards: Dean's List, Deputy Vice Chancellor's Scholarship, CMPS Undergraduate Award Nomination

Experience

Data Analyst Intern / Data POEM, Remote

May 2024 - Sep 2024

- Developed scalable connector modules using Mage.ai to interface with PostgreSQL and S3, facilitating the retrieval and transformation of website analytics data from multiple advertisement channels for machine learning model pipelines.
- Focused on data validation and verification to ensure the integrity of ML models, applying analytical skills to identify discrepancies.

Undergraduate Researcher / UBC Okanagan, Kelowna, BC

Sep 2023 - Jul 2024

- Conducted a study in HCI with 20+ participants evaluating cutting-edge command selection techniques using head-mounted displays, such as Hand-Proximate User Interface (HPUI) and gaze-tracked input. [Published](#) in ISS 2024.
- Explored the principles of In-Vehicle Infotainment Systems and built implementations of HPUI with C# and the Unity XR Toolkit.

Software Developer Intern / UBC Okanagan, Kelowna, BC

Sep 2023 - Dec 2023

- Spearheaded the development of a module within a graphical Decision Support System in Java for long-term degree planning, which is expected to help over 10,000 students plan courses and streamline discussions with academic advisors.
- Added support for over 120 complex degree configurations and used Google CP-SAT Solver to build an optimization model for optimal plan generation, along with various code refactoring and improvements to the optimizer.

Projects

Mossum: Weather Visualization macOS App / Swift, FastAPI, Redis, Render, SwiftData

May 2025 - Present

- Shipped a macOS app that displays live, customizable weather animations on the desktop using SpriteKit and the WeatherKit API.
- Designed a FastAPI backend with Redis caching to reduce latency and external API calls.
- Implemented versioned FAQ synchronization, enabling offline persistence in the app via SwiftData.
- Reached 2,700+ downloads and 30+ weeks in the top weather app charts on the Mac App Store.

FlightBar: Flight Tracking macOS App / Swift, FastAPI, Redis, Render

Feb 2025 - Present

- Built a flight tracker app backed by a FastAPI server that aggregates aviation, airport, airline, and weather data into a unified response.
- Implemented data validation, sanitization, and fallback logic to handle inconsistent data from external APIs.
- Added Redis caching with tiered TTLs (seconds up to days) to balance real-time updates, scheduled data, and long-lived assets.
- Currently with 800+ downloads on the Mac App Store.

Full-Stack E-Commerce Platform / PHP, MySQL, JavaScript, Stripe

Jan 2025 - Apr 2025

- Created a full-stack e-commerce website complete with user authentication, dynamic product listings, persistent carts, a responsive user interface, and order management using PHP and MySQL.
- Integrated Stripe checkout for secure payment flows and order placement.

Full-Stack AI-Powered Budgeting Website / Next.js, Plaid API, Pinecone, PostgreSQL, Docker

Sep 2024 - Apr 2025

- Developed a full-stack budgeting platform to help users visualize and manage spending habits using LLM-driven insights and a user context-aware chatbot powered by the Gemini API, applying software development lifecycle principles for smooth sprints.
- Engineered a vector-based transaction retrieval system by embedding and storing user financial data from Plaid API in Pinecone and PostgreSQL for fast, semantic querying.

Full-Stack E-Learning Platform / Flask, JavaScript, jQuery, MySQL, Selenium

Jan 2024 - Apr 2024

- Collaborated in developing a Flask-based full-stack e-learning platform, employing Agile methodologies, CI/CD practices, UML for system design, and Test-Driven Development (TDD) with Selenium for automated testing and deployment to ensure quality.
- Implemented core features, like user and course management, discussion portals, assignment submissions, and grading, with jQuery.

Skills

Languages: Swift, Python, JavaScript, TypeScript, Java

Frontend: React, Next.js, HTML, Tailwind CSS

Backend: FastAPI, Flask, Node.js, Express.js, PHP

Databases & ORMs: MySQL, PostgreSQL, Redis, Prisma, SQLAlchemy, SwiftData

Data & Pipelines: R, Pandas, NumPy, Seaborn, Pinecone, Mage.ai

Tooling: Git, Docker, Render, Netlify, Vercel